

Functional Requirements for TechReserve (Admin Side)

1. Authentication and Access Control

Module Capability	Detailed Specification
Account Management	• The system shall integrate with FEU Tech's Google Workspace via OAuth 2.0 to allow students, faculty, and staff to log in using their @feutech.edu.ph / @fit.edu.ph email addresses. • For manual logins, the system shall enforce a strict password policy (minimum 8 characters, alphanumeric, special characters) and account lockout after five failed attempts.
Role-Based Permissions	• The system shall enforce distinct access levels: • Borrower (Student/Faculty): Can submit requests, track status, and upload Activity Proposal Forms (APFs). • Admin (FO Manager/Coordinator): Full system access. Can review, approve, or reject reservations, configure optimization constraints, override task schedules, and extract analytical reports.
Session Management	• The system shall automatically terminate the admin session after 30 minutes of inactivity. • All login, logout, and session expiration events shall be logged with timestamp and user reference.

2. Admin Profile, Role, and Permission Management Module

Module Capability	Detailed Specification
Admin Profile Details	• The system shall maintain the following administrator profile attributes: Last Name varchar 50 First Name varchar 50 Email Address varchar 100 Role Designation enum (Admin) Contact Number varchar 20 Last Login Timestamp datetime

Account Controls	• The system shall allow authenticated administrators to: • View profile details • Update contact number • View role designation • Role assignment shall only be managed by FO Admin
Permissions	• Administrator access shall be restricted based on the assigned role. • Full system access • Manage equipment records • Manage user roles • Approve/reject reservations • Override readiness recommendations • Access analytics configuration

3. Equipment & Inventory Management Module

Module Capability	Detailed Specification
Equipment Registration	• The system shall allow FO Admin to create equipment records containing: • Equipment Name • Category • Total Quantity • Schedule • Operational Status (Active/Inactive)
Equipment State Management	• The system shall track equipment lifecycle states: • Available • Reserved • Approved • Prepared • Deployed • Returned • Under Maintenance • Each state transition shall record: • Timestamp • Administrator responsible

	• Reason (if applicable)
Inventory Visibility Control	• The system shall provide real-time visibility of equipment availability • The system shall prevent approval of reservations when available quantity is insufficient.

4. Reservation and Workflow Management

Module Capability	Detailed Specification
Reservation Review	• The system shall display all reservation requests in a structured queue. • Each reservation shall contain: • Reservation ID (Format: TR-YYYY-NNN) • Organization Name • Requested Venue • Requested Equipment • Quantity • Event Date and Time • Purpose/Activity Time • Submission Timestamp • Current Status • Reservations shall initially be marked as: • Pending Review
Unified Booking Interface	• The system shall mandate the upload of supporting documents for non-academic events and organizational academic events before submission can be finalized by the admin. The supporting documents are the following: • Approved Activity Proposal Form (APF) • Floor Plan • List of External Participants (if event requires external participants) All of which are uploaded in pdf file format.

Automated Priority Processing	• The system shall automatically tag and rank pending requests based on the institutional hierarchy: Academic (Classes/Thesis) > Institutional Events > Organizational Activities > Individual Use. • The system shall not automatically approve based on priority. Priority shall serve as a decision-support indicator only. • The system shall instantly alert the FO Coordinator of overlapping equipment requests and highlight the higher-priority activity.
Readiness Verification Workflow	• Upon request approval, the system shall automatically generate preparation checklists for FO Staff prior to the scheduled deployment time.
Reservation Decision Controls	• Admin shall be able to: • Approve • Reject • Request Revision • Rejected reservations shall require a mandatory reason entry. Approved reservations shall automatically update equipment status to “Approved”.

5. Release & Return Management Module

Module Capability	Detailed Specification
Release Processing	• Admin shall mark approved reservations as: • Prepared • Deployed • The system shall record: • Release timestamp • Assigned FO personal
Return Processing	• Admin shall confirm equipment return and record: • Return timestamp • Condition (Good / Damaged / Incomplete) • Notes/Remarks • Upon confirmation, equipment status shall automatically update to “Available”.

6. Predictive Analytics Engine (Time Series & Random Forest)

Module Capability	Detailed Specification
Peak Demand Forecasting	• The system shall utilize a Time Series Forecasting model analyzing historical reservation logs to predict high-demand periods (e.g., midterm/finals week). • The dashboard shall display a 7-day and 30-day visual forecast of expected inventory depletion for critical assets (e.g., mics, sound systems).
Readiness Risk Scoring	• The system shall process every pending reservation through a Random Forest machine learning model to generate a "Readiness Risk Score" (0% - 100%). • The model shall evaluate variables such as turnaround time between consecutive bookings, equipment fragility, and historical delay rates. • The system shall automatically flag high-risk reservations (e.g., >75% risk) on the FO Coordinator's dashboard for preemptive action.

7. Prescriptive Optimization (Linear Programming)

Module Capability	Detailed Specification
Auto Task Generation	• Upon approval and daily lock-in of reservations, the system shall automatically generate structured operational tasks for FO Staff. • Tasks shall be categorized as: – Deployment Task – Monitoring Task – Retrieval Task – Emergency Response Task (if triggered by risk flag or equipment issue). • Each auto-generated task shall contain the following attributes: – Task ID (Format: TS-YYYYMMDD-NNN) – Linked Reservation ID – Task Type – Assigned Equipment List – Venue Location – Start Time – End Time

	– Estimated Duration – Priority Level (Normal / High / Critical) – Readiness Risk Indicator (if applicable) – Assigned FO Staff – Task Status (Pending / In Progress / Completed / Reassigned)
Intelligent Task Designation	• The system shall automatically designate tasks to FO Staff using the Linear Programming optimization output. • Task assignment shall consider the following variables: – Staff availability and shift schedule – Break schedules – Staff workload balance – Required technical specialization (e.g., LED wall operator) – Venue proximity to reduce transit time – Readiness Risk Score prioritization • The objective function shall minimize total idle time and travel distance while ensuring all high-priority tasks are covered.
Dynamic Reassignment Logic	• If a staff member becomes unavailable (absence, emergency, delay), the system shall: – Automatically flag affected tasks – Re-run the optimization model in real-time – Generate a reassigned optimized schedule within 10 seconds • FO Coordinators shall receive immediate notification of auto-reassignments.
Task Conflict Prevention	• The system shall prevent overlapping task assignments for the same FO Staff. • The system shall validate that task duration does not violate shift or break constraints.
Manual Override Control	• FO Coordinators shall retain the ability to manually reassign or modify tasks through drag-and-drop interface. • Any manual override shall: – Log timestamp – Record Admin ID – Store override reason for audit tracking.
Audit & Traceability	• All auto-generated tasks and system-driven assignments shall be logged in an audit trail. • The system shall retain historical task allocation data for at least 5 years to support optimization model retraining and performance analysis.

8. Inventory and Asset Monitoring

Module Capability	Detailed Specification
Dynamic Asset Tracking	• The system shall track the real-time state of every registered asset: Available, Pencil-Booked, Approved/Reserved, Deployed, Pending Retrieval, or Under Maintenance.
Maintenance Ticketing	• FO Staff shall be able to flag an item as defective directly from their task dashboard upon retrieval. • The system shall automatically remove defective items from the "Available" pool and prompt the FO Coordinator to initiate a repair workflow.

9. Dashboards, Reporting, and Notifications

Module Capability	Detailed Specification
Centralized Analytics Dashboard	• The system shall provide a real-time visual dashboard displaying active deployments, total pending requests, equipment utilization rates, overdue returns, active readiness risk alerts.
Automated Alerts	• Admin shall receive notifications for: New reservation submission, reservation conflict detected, high readiness risk, upcoming peak demand period, overdue equipment return. • The system shall dispatch automated email and in-app notifications to requesters regarding status changes (e.g., Approved, Rejected, Missing Documents). • The system shall alert FO Staff 30 minutes prior to an optimized deployment or retrieval task.
Data Exportation	• The system shall generate and export (PDF/Excel) historical reports on equipment utilization, frequent scheduling conflicts, and predictive vs. actual demand accuracy.

Non-Functional Requirements for TechReserve

Category	Detailed Specification
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Performance	• Core database queries (e.g., checking equipment availability) shall execute in under 1.5 seconds. • The Random Forest readiness risk calculation shall process and display within 3 seconds of opening a pending reservation. • The Linear Programming optimization algorithm shall generate a complete daily staff schedule in under 15 seconds. • The web platform must support up to 500 concurrent student/faculty users and 50 concurrent FO personnel without noticeable latency.
Usability	• The system shall feature a responsive web design that adapts seamlessly to desktop monitors • The predictive analytics outputs (Risk Scores, Demand Forecasts) must be presented with intuitive, color-coded visual indicators (e.g., Red/Yellow/Green) to ensure non-technical staff can easily interpret the data.
Reliability and Availability	• The system shall maintain an uptime of 99.5% during critical university operating hours (6:00 AM to 10:00 PM). • Graceful Degradation: In the event that the predictive or prescriptive analytics modules experience a critical failure, the core reservation, inventory, and manual-approval workflows must remain fully operational.
Security	• All data transmissions must be encrypted using TLS/SSL (HTTPS) protocols. • The system shall comply with the Philippine Data Privacy Act of 2012 (RA 10173), ensuring that user data (student IDs, contact information) is securely encrypted at rest. • Database backups shall be executed automatically every 24 hours, utilizing a 30-day rolling retention policy stored on secure cloud infrastructure.
Scalability	• The database architecture shall be designed to efficiently index and store up to 5 years of detailed transactional reservation logs, as this volume of historical data is critical for continuously training and refining the Time Series forecasting model.